

A Case Study on Student Placement Prediction

Data Science and Analytics (CMP 360)

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Outline

- 1 Introduction
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- 3 Data Visualization
- 4 Statistical Analysis
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Problem Statement

- Campus placement outcomes matter to students, institutions, and recruiters alike.
- Goal: understand which student attributes — demographic (age, gender, city) or academic (CGPA, degree level) — are meaningfully associated with placement.
- Dataset: **1,100 student records** with demographic and academic attributes plus a binary placement outcome.

Central Question

“Which student attributes, if any, are statistically associated with placement status?”

- Workflow: data cleaning → transformation → encoding → outlier handling → visualization → hypothesis testing.

Information about the Data

Column	Type	Description	Missing
name	Categorical (text)	Student's first name	85 (7.7%)
city	Categorical (nominal)	City of residence	184 (16.7%)
gender	Categorical (binary)	male / female	99 (9.0%)
profession	Categorical (ordinal)	Highest degree (bachelor/masters/phd)	169 (15.4%)
age	Numeric (continuous)	Student's age in years	112 (10.2%)
cgpa	Numeric (continuous)	CGPA (0–10 scale)	132 (12.0%)
placed	Binary (target)	1 = placed, 0 = not placed	0 (0%)

Table: Dataset column summary — 1,100 rows $\times$ 7 columns

Dataset Shape & Sample

Shape: (1100, 7)

(rows, columns)

name	city	gender	profession	age	cgpa	placed
moriarity	NaN	female	phd	28.0	5.94	1
moriarity	asgard	NaN	bachelor	50.0	8.55	0
holmes	NaN	female	masters	18.0	5.56	0
sam	NaN	male	bachelor	25.0	8.57	1
sam	NaN	male	bachelor	19.0	8.76	1

`df.head()` output

- Target variable `placed` is well balanced: **521 not placed (50.3%)** vs. **514 placed (49.7%)**.
- A near 50/50 split is favorable for unbiased group comparisons.

Dataset Info & Missing Values

#	Column	Non-Null Count	Dtype
0	name	1009 non-null	str
1	city	905 non-null	str
2	gender	994 non-null	str
3	profession	921 non-null	str
4	age	982 non-null	float64
5	cgpa	962 non-null	float64
6	placed	1100 non-null	int64

`df.info()` — RangeIndex: 1100 entries, 0 to 1099

Memory usage: 60.3 KB **Dtypes:** float64(2), int64(1), str(4)

Source of Data

- **Dataset:** `my_data.csv` — sourced from Kaggle, "*All-in-One Dataset for Preprocessing Practice*" by Akshat Sharma.¹
- **Stated purpose:** synthetically generated via custom Python code to serve as an all-in-one resource for practicing preprocessing techniques (missing values, outliers, duplicates, scaling, encoding).
- **Not** sourced from a real placement cell — it *simulates* student placement outcomes based on degree, age, and CGPA.
- **Notable design choice:** names and cities are deliberately drawn from fictional universes (Sherlock Holmes, Supernatural, MCU, DCU) — an intentional touch by the author, not a data quality flaw.

¹[https:](https://www.kaggle.com/datasets/akshatsharma2/all-in-one-dataset-for-preprocessing-practice/data)

[//www.kaggle.com/datasets/akshatsharma2/all-in-one-dataset-for-preprocessing-practice/data](https://www.kaggle.com/datasets/akshatsharma2/all-in-one-dataset-for-preprocessing-practice/data)

Source of Data: Column Descriptions

Column	Type	Description
name	Categorical	Fictional character names (Sherlock, Watson, Sam, Dean, etc.)
city	Nominal	Fictional locations (Gotham, Wakanda, Asgard, Purgatory)
gender	Nominal	Male / Female
profession	Ordinal	Educational qualification: Bachelor < Master < PhD
age	Numerical	Age in years
cgpa	Numerical	Cumulative Grade Point Average
placed	Label	Placement status (target variable)

Column descriptions per dataset documentation

Implication: findings characterize *this dataset and methodology*, not a real student population.

Data Cleaning: Duplicate Records

- Checked with `df.duplicated().sum()`.
- Raw dataset ($1,100 \times 7$) contained **65 fully duplicate rows (5.9%)**.
- Removed via `df.drop_duplicates()`.
- Post-cleaning shape: **(1,035, 7)** — verified with a second `df.duplicated().sum()` call returning **0**.

Stage	Rows	Columns	Duplicates
Before cleaning	1,100	7	65
After cleaning	1,035	7	0

Duplicate removal summary

Data Cleaning: Sample Duplicate Rows

name	city	gender	profession	age	cgpa	placed
castiel	asgard	male	masters	24.0	7.71	1
NaN	wakanda	male	bachelor	27.0	7.90	0
moriarity	asgard	male	phd	25.0	NaN	1
sherlock	gotham	male	masters	28.0	7.55	1
dean	wakanda	male	bachelor	20.0	9.31	0
			⋮			
moriarity	gotham	male	NaN	26.0	7.96	0
holmes	NaN	male	masters	26.0	8.92	1
moriarity	wakanda	male	masters	19.0	9.01	0
sam	asgard	male	bachelor	30.0	7.88	0
sam	asgard	NaN	masters	24.0	8.94	1

Excerpt of the 65 detected duplicate rows (`df[df.duplicated()]`)

Note: rows 1000–1099 correspond to duplicate indices in the original DataFrame; only a subset is shown here for brevity.

Missing Values: Detection

- Checked with `df.isnull().sum()` — all 7 non-target columns affected (target placed clean).
- Missing proportions ranged **8%–18%**; rows were **not** dropped outright to preserve sample size.

Column	Missing Count	Missing %
name	85	8.21%
city	184	17.78%
gender	99	9.57%
profession	169	16.33%
age	112	10.82%
cgpa	132	12.75%
placed	0	0.00%

Missing values by column (`df.isnull().sum()`)

Missing Values: Imputation Strategy & Type Fix

Column(s)	Type	Strategy
age, cgpa	Numeric	Filled with column median (robust to skew/outliers)
city, gender, profession	Categorical	Filled with column mode
name	Categorical	Filled with placeholder "Unknown"

Imputation strategy by column type

- Verified with `df.isnull().sum()` → **0 missing** across all columns.
- age was stored as `float64` despite representing whole years → cast to `int` via `df["age"].astype(int)`.

Data Standardization

- age and cgpa standardized using scikit-learn's StandardScaler (z-scores).
- Necessary before any distance- or scale-sensitive analysis.

name	age	cgpa	age_std	cgpa_std
moriarity	28	5.94	0.035	-1.169
moriarity	42	8.55	1.885	0.662
holmes	18	5.56	-1.287	-1.435
sam	25	8.57	-0.362	0.676
sam	19	8.76	-1.155	0.809

Sample rows: original vs. standardized age and cgpa

Metric	age_std	cgpa_std
Mean	≈ 0	≈ 0
Std. Dev. (ddof = 0)	1.0	1.0

Verification of standardization

Data Transformation: Binning

Age_Group

Bin	Range	Count
Young	18–25	437
Adult	25–35	398
Senior	35–60	200

CGPA_Category

Bin	Range	Count
Low	4–6	200
Average	6–8	412
High	8–10	423

age	Age_Group	cgpa	CGPA_Category
28	Adult	5.94	Low
42	Senior	8.55	High
18	Young	5.56	Low
25	Young	8.57	High
19	Young	8.76	High

Bins via `pd.cut()` with `include_lowest=True`

Handling Categorical Data: Encoding

- A readable copy (`df_viz`) was preserved **before** encoding, for use in visualization and hypothesis testing.
- **Label Encoding** on gender (binary): female $\rightarrow$ 0, male $\rightarrow$ 1.
- **One-Hot Encoding** on city and profession (`pd.get_dummies`) — avoids implying a false ordinal relationship in nominal variables.

name	gender (encoded)	Original
moriarity	0	female
moriarity	1	male
holmes	0	female
sam	1	male
sam	1	male

Label encoding applied to gender via `LabelEncoder`

Encoding: One-Hot Result (city, profession)

name	city_asgard	city_gotham	city_purgatory	city_wakanda
moriarity	0	0	0	1
moriarity	1	0	0	0
holmes	0	0	0	1
sam	0	0	0	1
sam	0	0	0	1

name	profession_bachelor	profession_masters	profession_phd
moriarity	0	0	1
moriarity	1	0	0
holmes	0	1	0
sam	1	0	0
sam	1	0	0

Original city and profession columns replaced by binary indicator columns

Outlier Detection: IQR Method

- Two complementary methods: **box plots** (visual) and the **IQR rule**.
- Bounds: values $< Q1 - 1.5 \times IQR$ or $> Q3 + 1.5 \times IQR$ flagged as outliers.

Variable	Lower Bound	Upper Bound
age	10.0	42.0
cgpa	– ~5.2 (low-end cluster)	

IQR bounds computed via $Q1 - 1.5 \times IQR$ / $Q3 + 1.5 \times IQR$

- **Age:** 154 rows flagged (high-end, above ~42).
- **CGPA:** cluster of low-end outliers below ~5.2.

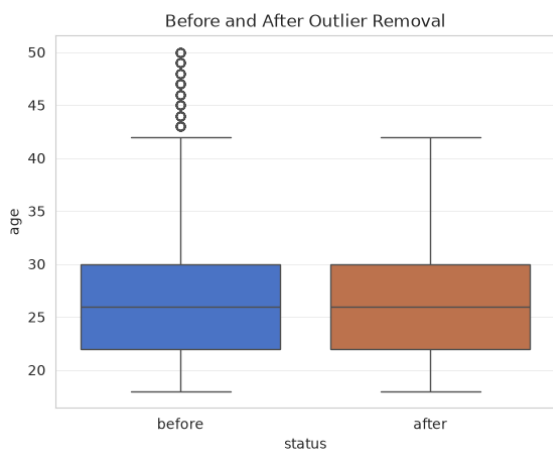
Outlier Detection: Sample Flagged Rows & Clipping

name	city	profession	age	cgpa
moriarity	asgard	bachelor	50	8.55
sherlock	wakanda	bachelor	48	8.75
moriarity	wakanda	masters	43	7.74
sam	gotham	bachelor	44	9.30
dean	wakanda	masters	46	7.89
		⋮		
dean	gotham	bachelor	43	5.00
sam	wakanda	bachelor	47	7.76
sherlock	gotham	bachelor	44	8.86

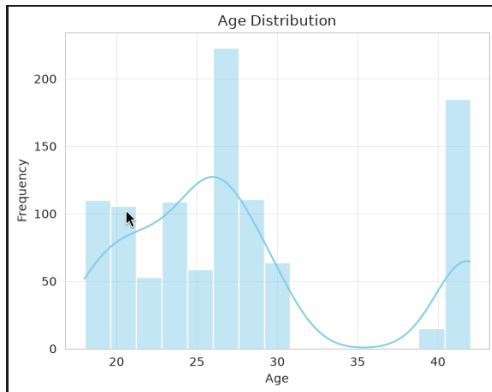
Excerpt of 154 rows flagged via age IQR bounds

- Rather than deleting ($\sim 15\%$ data loss), extreme values were **clipped** to IQR boundaries via `df["age"].clip(lower_limit, upper_limit)`.
- Zero rows remained out-of-bounds afterward.

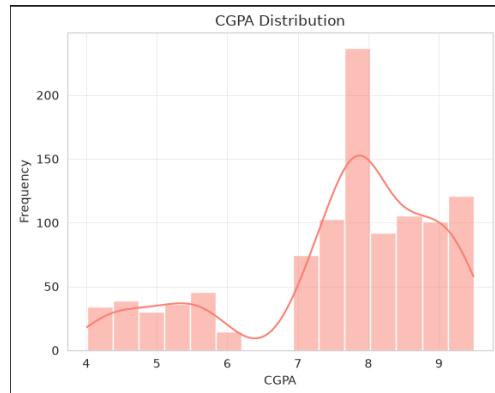
Outlier Clipping: Before vs. After



Univariate Distributions: Age & CGPA



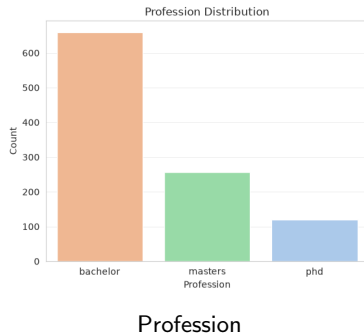
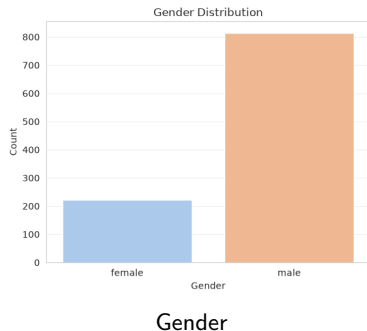
Age Distribution



CGPA Distribution

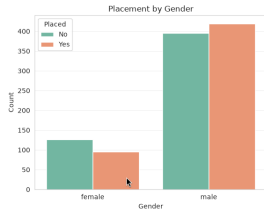
- **Age:** mildly bimodal, with a secondary cluster around 40–42 years.
- **CGPA:** right-skewed, with a secondary low-scoring cluster below 6.

Univariate Distributions: Gender, City, Profession

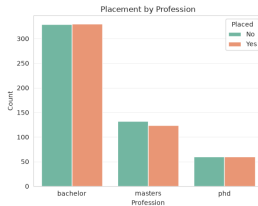


- **Gender:** male (~813) considerably outnumber female (~220) students.
- **City:** Wakanda dominant (~547), followed by Gotham, Asgard, Purgatory.
- **Profession:** Bachelor's dominant (~657), then Master's (~254), PhD (~119).

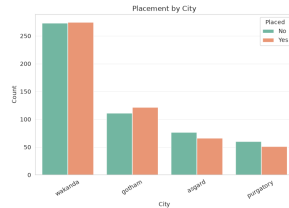
Placement vs. Categorical Features



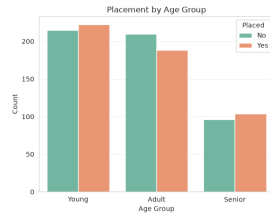
Gender



Profession



City



Age Group

Placement Rate by Category

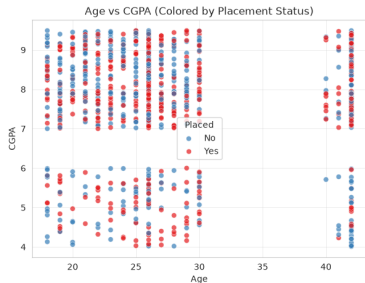
Gender	Placement Rate
Female	42.99%
Male	51.47%

Profession	Placement Rate
Bachelor	50.08%
Masters	48.44%
PhD	50.00%

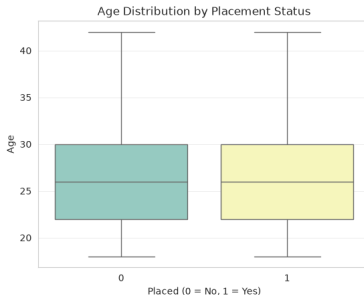
City	Placement Rate
Asgard	46.15%
Gotham	52.36%
Purgatory	45.95%
Wakanda	50.18%

Gender shows the widest gap (male 51.5% vs. female 43.0%) — foreshadows the significant chi-square result.

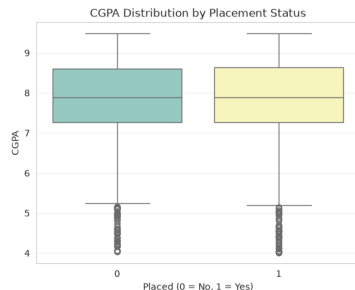
Placement vs. Numeric Features



Age vs. CGPA Scatter



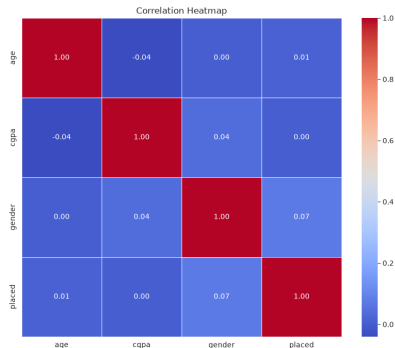
Age by Placement



CGPA by Placement

- No visible separation between placed and not-placed groups on the Age vs. CGPA scatter plot.
- Box plots of Age and CGPA by placement status show **near-identical medians and spread**.

Correlation Heatmap



- All pairwise correlations are very close to zero (largest magnitude: 0.07, between gender and placed).
- Visually confirms the absence of any strong linear relationship among these variables.

Hypothesis Testing: Independent t -tests

Test	H_0	t	p	Result
CGPA vs. Placed	No difference in mean CGPA	0.1197	0.9048	Fail to reject
Age vs. Placed	No difference in mean Age	0.2921	0.7703	Fail to reject

- Neither CGPA nor Age shows a statistically significant difference between placed and not-placed students ($p \gg 0.05$).
- Consistent with the near-identical medians/spread seen in the earlier box plots and the near-zero correlations.

Hypothesis Testing: Chi-square Tests

Test	H_0	χ^2	df	p	Result
Gender vs. Placed	Independent	4.6751	1	0.0306	Reject
Profession vs. Placed	Independent	0.2042	2	0.9029	Fail to reject

Gender	Not Placed	Placed
Female	126	95
Male	395	419

- Gender shows a **significant association** with placement ($p = 0.0306 < 0.05$).
- Profession shows **no significant association** with placement ($p = 0.9029$).

Correlation Analysis

	age	cgpa	gender	placed
age	1.000	−0.040	0.003	0.009
cgpa	−0.040	1.000	0.037	0.004
gender	0.003	0.037	1.000	0.070
placed	0.009	0.004	0.070	1.000

Pair	Pearson r	p -value	Significant?
Age vs CGPA	−0.0404	0.1945	No
Age vs Placed	0.0091	0.7703	No
CGPA vs Placed	0.0037	0.9048	No
Gender vs Placed	0.070	–	Consistent w/ χ^2 result

Key Findings

- **Gender** is the only feature with a statistically significant association with placement ($\chi^2 = 4.6751$, $p = 0.0306$): male 51.5% vs. female 43.0% (~ 8.5 pp gap).
- **CGPA** shows no significant relationship ($p = 0.9048$); placed vs. not-placed means are nearly identical (7.61 vs. 7.60).
- **Age** shows no significant relationship ($p = 0.7703$).
- **Profession** shows no significant association ($p = 0.9029$); placement rates ~ 48 – 50% across all degree levels.
- **City** shows a modest raw spread (45.9%–52.4%) but was not formally tested — likely sampling noise.
- **Age and CGPA** are themselves uncorrelated ($r = -0.0404$).

- Placement outcome behaves close to a **random 50/50 split** with respect to CGPA, age, and degree level.
- The only detectable (and still modest) signal is a **gender-based difference**.
- This is counter-intuitive relative to the common assumption that CGPA strongly predicts placement — worth flagging explicitly.
- Given the fictional names/cities, placement outcomes were plausibly generated by a **random or near-random process** (possibly with a small deliberate gender effect).

Caveat

Conclusions are valid descriptions of *this dataset and methodology* — they should not be over-generalized to real placement processes without validation on real institutional data.

Challenges & Future Enhancements

Challenges

- High missing-value rates (8%–18%) required column-specific imputation, not row deletion.
- 5.9% exact duplicate rows identified and removed.
- Skewed distributions required IQR clipping over naive z-score filtering.
- Weak/non-significant correlations $\Rightarrow$ a clean “no strong effect” is still a valid finding.
- Encoding choices (label vs. one-hot) made carefully to avoid false ordinal relationships.

Future Enhancements

- Richer features: interviews, skill assessments, internships, extracurriculars.
- Predictive ML models (Logistic Regression, Random Forest, Gradient Boosting).
- Multivariate/interaction analysis (e.g., two-way ANOVA).
- Validate on a real institutional placement dataset.
- Advanced imputation (KNN, MICE).
- Interactive dashboard (Power BI, Tableau, Streamlit).